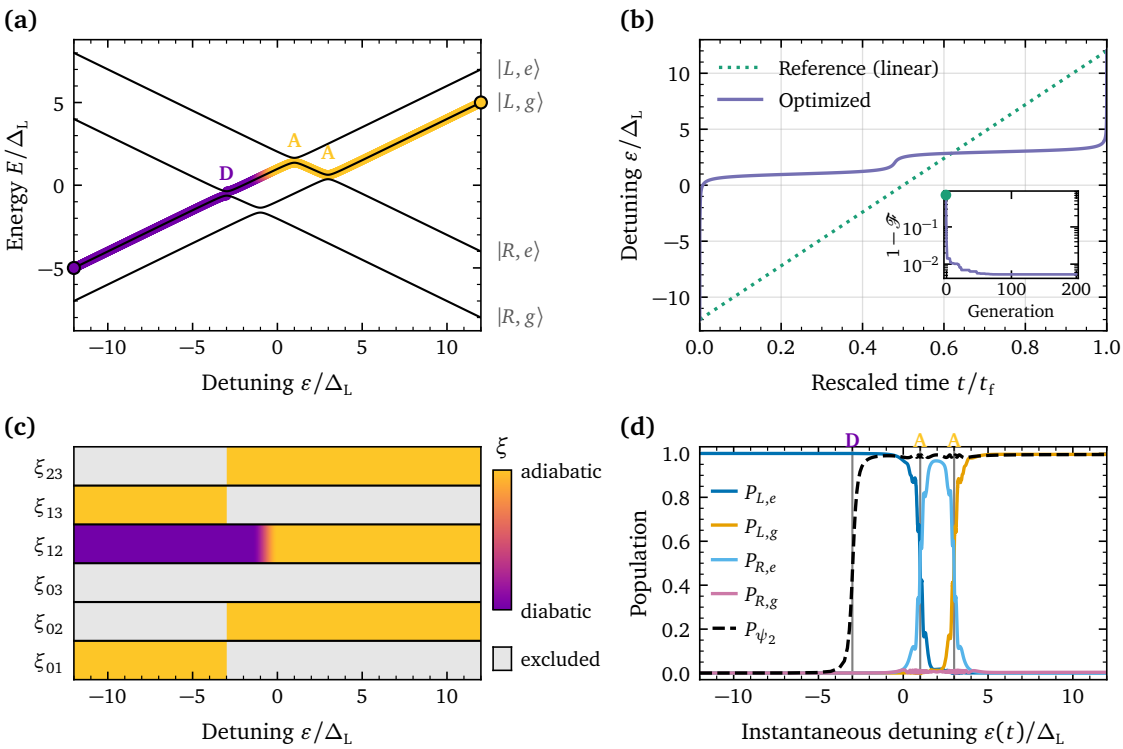

(a)
Energy E/Δ_L
Detuning ε/Δ_L
$|L,e\rangle$
$|L,g\rangle$
$|R,e\rangle$
$|R,g\rangle$
D
A
A
(b)
Detuning ε/Δ_L
Rescaled time t/t_f
Reference (linear)
Optimized
$1-\mathcal{F}$
Generation
(c)
ξ_{23}
ξ_{13}
ξ_{12}
ξ_{03}
ξ_{02}
ξ_{01}
ξ
adiabatic
diabatic
excluded
Detuning ε/Δ_L
(d)
Population
$P_{L,e}$
$P_{L,g}$
$P_{R,e}$
$P_{R,g}$
P_{ψ_2}
D
A
A
Instantaneous detuning $\varepsilon(t)/\Delta_L$